MEMORANDUM

TO: Dr. Tina Han

FROM: Hengeltraub, Isabella; Wu, Xintong; Jhanwar, Yuvraj; Hassan, Khaleel

CC: ME32301 Lab Section 08- Harbin, Zachary; Alshaikh, Leen

DATE: April-04-2025

RE: Uniaxial Tension in a Finite-Width Panel with Circular Hole: Lab 8

EXECUTIVE SUMMARY

The goal of this lab is to determine the stress concentration factor (SCF), given by K, for a finite-width panel of various hole sizes and mesh configurations that is subjected to uniaxial tension. The stress concentration factor K is a dimensionless quantity that expresses the intensity of local stress compared to the stress calculated by presuming a uniform distribution, known as nominal stress σ_{nom} [1]. K is given by

$$K = \frac{\sigma_{max}}{\sigma_{nom}} \tag{1}$$

K_{theo} is calculated using

$$K = 3 - 3.14(\frac{2r}{D}) + 3.667(\frac{2r}{D})^2 - 1.527(\frac{2r}{D})^3 \tag{2}$$

σ_{nom} is given by

$$\sigma_{nom} = \frac{P}{(D-2r)t} \tag{3}$$

where P is the external load, D is the width of the finite-width panel, and $2r$ is the diameter of the hole, and $2r/D$ is the ratio of the hole diameter to the panel width. The experiment is performed in a virtual Finite Element Analysis (FEA) environment. In the simulation, the finite-width panel is made of generic steel with dimensions D = 180 mm and t = 10 mm. The panel is fixed on one end and a load of P = 100 N is applied on the other end. The data was manually recorded 72 times,

testing eight different hole sizes $2r/D$ of 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, and 0.8. Each hole size was examined using nine different mesh sizes ranging from 1 mm to 9 mm in order to analyze the effect of varying hole and mesh configurations on stress concentration in a finite-width panel. It is important to note that, the smaller the mesh size, the more accurate the simulation results ought to be. From each hole and mesh size, σ_{max} was recorded manually in an excel document. The max stress on the simulation was typically located at the outermost part of the hole, corresponding to the most intense red region. Using equation 3 and 1, the stress concentration factor K for each hole and mesh size is determined. K_{FEA} is then compared to K_{theo} theoretical and the percentage difference is calculated. Table 1 shows a summary of the results. It is evident that as mesh size increases, the %diff between K_{theo} and K_{FEA} increases, as a result of reduced mesh accuracy. However, an exception is observed in the final case, likely due to potential errors such as inconsistencies in selecting the absolute maximum stress value on the panel. Furthermore, σ_{max} for hole and mesh sizes configurations (0.1, 2mm), (0.1, 5mm), (0.2, 2mm) did not occur around the perimeter of the hole. This is likely due to the small hole sizes relative to the mesh resolution, preventing accurate stress analysis of the panel. Equally important, while the summary data reflects the correct overall trend, the maximum stress values recorded during the experiment fluctuated significantly. These fluctuations introduced variations in the %diff, even in cases where the mesh sizes were more accurate. These inconsistencies may stem from the manual method used to identify σ_{max}, which relied on visual inspection and cursor selection, giving rise to potential human error. One way this can be improved could be to automate the data collection process and use a computer to obtain the max stress value in the data set. Additionally, there might have been errors due to data processing and inputting the values as well.

$2r/D$	σ_{nom} [MPa]	L_{mesh} [mm]	σ_{max} [MPa]	K_{FEA}	K_{theo}	% diff
0.1	0.0617	1	0.1504	2.44	2.72	10.4
0.1	0.0617	3	0.1272	2.06	2.72	24.3
0.1	0.0617	5	0.1066	1.73	2.72	36.5
0.1	0.0617	7	0.0937	1.52	2.72	44.2
0.1	0.0617	9	0.1314	2.13	2.72	21.8

Table 1: Comparison of theoretical and experimental stress concentration factors.

RESULTS

Table 2 shows the % difference between the theoretical and experimental stress concentration factors, with $2r/D$ being the ratio of the diameter $2r$ of the hole and the width D of the panel, σ_{nom} being the nominal stress, L_{mesh} being the mesh size, σ_{max} being the maximum stress, K_{FEA} being the experimental stress concentration factor, K_{theo} being the theoretical stress concentration factor, and % diff being the percent difference. σ_{nom} is calculated using Equation (3), with P being external stress and t being the thickness of the panel. K_{FEA} is calculated using Equation (1), and K_{theo} is calculated using Equation (2).

$2r/D$	σ_{nom} [MPa]	L_{mesh} [mm]	σ_{max} [MPa]	K_{FEA}	K_{theo}	% diff
0.1	0.0617	1	0.1504	2.44	2.72	10.4
0.1	0.0617	3	0.1272	2.06	2.72	24.3
0.1	0.0617	5	0.1066	1.73	2.72	36.5
0.1	0.0617	7	0.0937	1.52	2.72	44.2
0.1	0.0617	9	0.1314	2.13	2.72	21.8

Table 2: Comparison of theoretical and experimental stress concentration factors.

As shown in the table above, the % difference between the theoretical and experimental stress concentration factors generally increase as mesh size increases, which is expected, as a larger mesh size means the measurement is less precise.

The % difference between the theoretical nominal stress and the nominal stress from the simulation is calculated using the simulation result of $2r/D = 0.1$ and a mesh size of 5 mm, as seen in Figure 1. Taking the value of 0.062545 MPa from the figure as the experimental value and 0.0617 as the theoretical value, as shown in Table 2, the % difference is 1.37%.

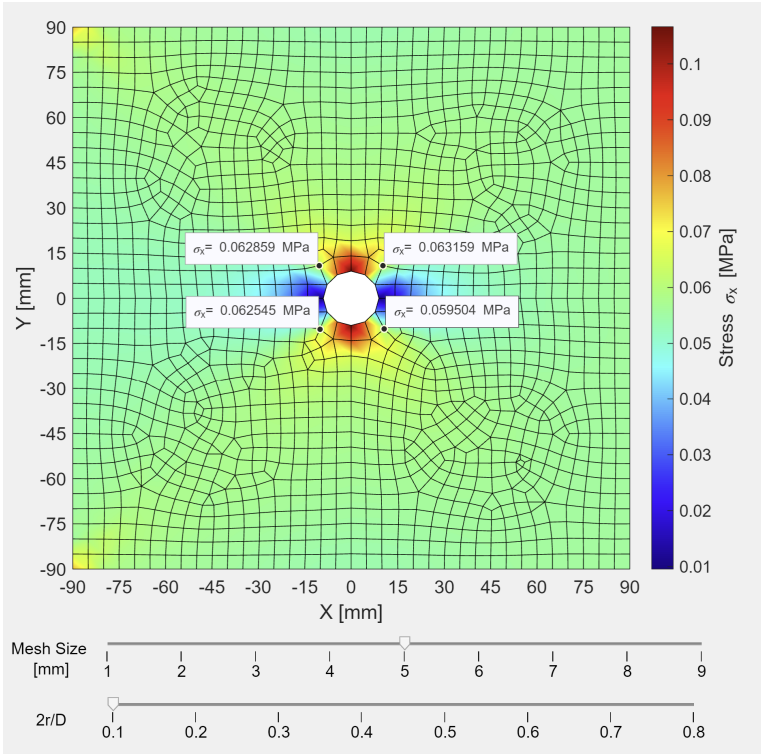

Figure 1: σ_{nom} values from simulation

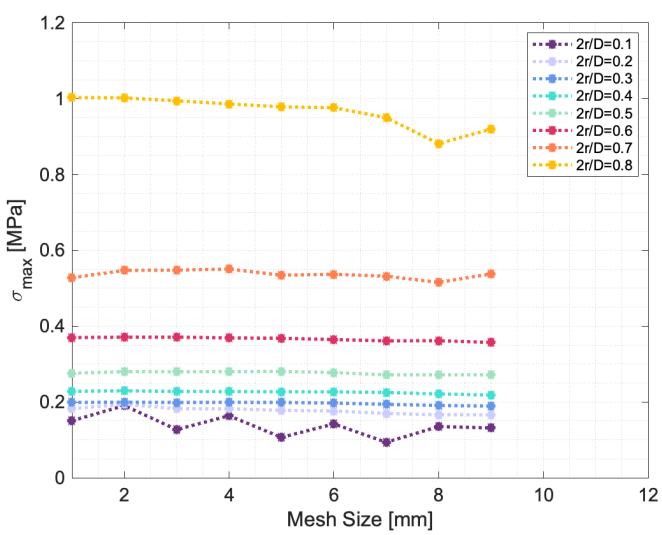

Figure 2: σ_{max} values compared to mesh size for each hole size

In Figure 2, maximum stress is plotted against mesh size for each hole size. As shown in the figure, maximum stress remains fairly consistent as mesh size changes, although there is more variation for the smallest hole. This is expected because even the smallest mesh size is too coarse for the

small hole, leading to more inaccurate results. Additionally, maximum stress is higher for bigger holes, which makes sense because the force is spread out over a smaller area.

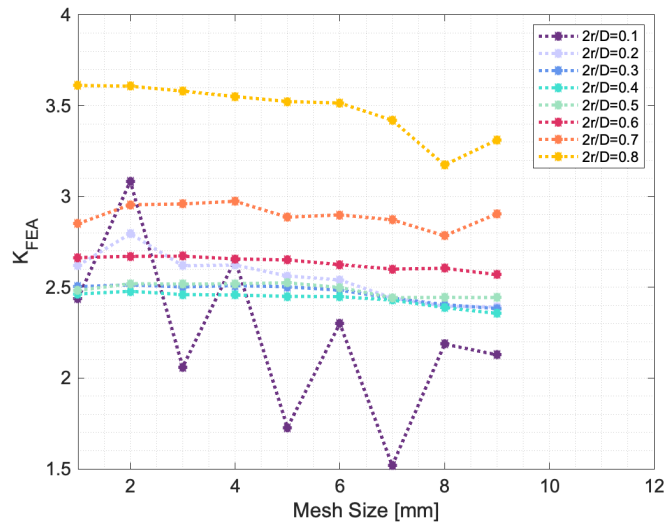

Figure 3: K_{FEA} values compared to mesh size for each hole size

In Figure 3, the stress concentration factor is plotted against mesh size for each hole size. Again, the values are fairly consistent for all mesh sizes except for the smallest hole size, which makes sense because the stress concentration factor depends only on nominal stress, which is constant for across mesh sizes, and maximum stress is varied, as explained above.

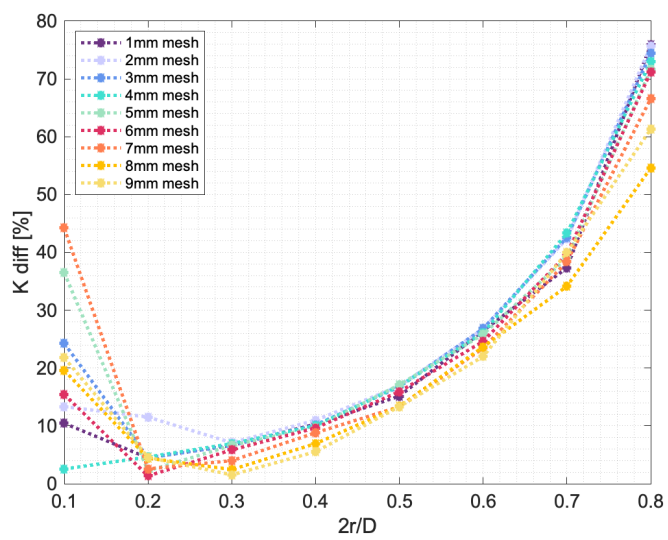

Figure 4: K % difference compared to hole sizes for each mesh value

In Figure 4, the % difference between the theoretical and experimental stress concentration factor

is plotted against the hole sizes. The % difference increases as the hole size increases because it is harder to pinpoint where the maximum stress is for larger hole sizes. The exception is the smallest hole with a $2r/D$ of 0.1, which, as stated before, is too small even for the finest mesh size to measure accurately.

The MATLAB code for the graphs was done using an iterative process. Figures 2 and 3, for σ_{max} and K_{FEA}, were done very similarly. The data used to create the code was organized into 8 sections, one for each hole size from 0.1 to 0.8, with a 0.1 increment. Within those sections were nine mesh sizes, ranging in order from from 1 mm to 9 mm, with a 1 mm increment. For all of those mesh size and hole combinations, the σ_{max} and K_{FEA} were recorded. Figures 2 and 3 were created with a "for" loop, which went through each mesh size within a hole size, and assigned those vales to a graph representing either σ_{max} (Figure 2) or K_{FEA} (Figure 3). As such, it then had eight graph colors, one color for each hole size (2r/D) value. The code first took the nine consecutive mesh values (1–9 mm) for the first hole size (0.1), then the next nine for the 0.2 hole size, and so on for a total of eight iterations.

Figure 4 was created a little bit differently, as it plotted each mesh size as its own colored graph, making for nine different graph colors, with the eight hole sizes now making up the x-axis. The percent difference in the K values between calculated and theoretical values was plotted on the y-axis. Much like Figures 2 and 3, this was done with an iterative "for" loop, but instead of taking nine consecutive value readings (which is 1 mm to 9 mm mesh size readings for a single hole size), it had to look at all the mesh values for each hole size. In order to do this, the code had a total of eight iterations, starting at the first hole reading for that size hole and then skipping eight rows to arrive at the next same mesh value for the next hole size until the end of the code. All graphs were created using the same color list, with Figure 4 having one additional color (the lightest yellow) to create a rainbow aesthetic. They were also given appropriate letter formatting and axis bounds for clarity.

CONCLUSIONS

The purpose of this lab was to calculate the value of K such that a trend can be observed when different meshes are used for different circular hole sizes. This goal was met by investigating eight different circular hole sizes with nine different mesh sizes to examine and simulate the stress concentration in a finite-width panel undergoing uniaxial tensile loading. The mesh size varied from 1 mm to 9 mm, where 1 mm was the smallest mesh size and 9 mm was the largest mesh size.

Based on results from Table 2, it can be noted that for a small hole size of 0.1, the nominal stress σ_{nom} remains constant at 0.0617 MPa, while the σ_{max} progressively decreases from 0.1504 MPa to 0.1314 MPa as mesh size was increased from 1 mm to 9 mm. This shows a larger deviation from theoretical values, and thus causes higher percentage differences as the mesh size increases. It can be inferred from Figure 2 that the σ_{max} value remains fairly constant for most holes except the smallest hole (0.1) and the largest hole (0.8). Moreover, in Figure 3, a similar trend can be noticed, with all other holes showing consistent K_{FEA} values whereas the smallest hole (0.1) showing largest deviation, followed by the 0.8 hole.

In theory, as mesh size decreases, the maximum stress captured is more accurate. This is because finer meshes allow for the model to capture the stress gradients more closely and perform interpolation with more accurate values. The experimental results documented in Table (2) depict a general trend that with increasing mesh sizes, the percentage difference in the value of K increased. For instance, the 1 mm mesh had 10.5 % difference, the 3 mm mesh had 24.3 % difference, the 5 mm mesh had 36.5 % difference and the 7 mm mesh had 44.3 % difference. However, the percentage difference was considerably small for the largest mesh, the 9 mm mesh, at 21.8 % difference. After further analysis of all the collected data, it was noticed that as the hole size was increased, increasing mesh size would reduce percentage error in the value of K. Thus, for smaller holes the finer mesh sizes would improve accuracy and serve as a better mechanism to calculate K, because the stress gradient is sharper for smaller holes and requires a finer mesh to resolve. However, for bigger holes, larger mesh sizes perform well because for larger holes, the stress is not as localized

and thus larger meshes are able to capture the stress behavior effectively.

There was a clear trend upon looking at the variation of percentage difference in K and the hole sizes. As the hole size increased from 0.2 to 0.8, the percent error increased in general. For instance, the percent error in K when hole size was at 0.2 was about 2% on average, for 0.3 it was about 5.3%, for 0.7 it was about 40% and for 0.8 hole size it was about 69%. Thus, from hole size 0.2 to 0.8, there was a clear increase in error. This can be attributed to manual error because when the hole size increases, it is more difficult to locate the point where the stress is maximum, near the rim of the hole based on the stress concentration theory. However, this trend cannot be generalized for the 0.1 hole size because that is an exceptional case where the hole size is extremely small and for the range of mesh sizes 0.1 mm to 0.9 mm, the results are unreliable. Thus, the percentage differences for the 0.1 hole size do not give much insight into the actual error that could exist. This can be evidenced in Figure 4, wherein the trends in percentage difference show a somewhat quadratic relationship from hole sizes 0.2 to 0.8, however; for the 0.1 hole, the trend is different and is somewhat linear.

The principles of stress concentration and the Finite Element Analysis Method are important tools in various engineering disciplines. For instance, in the aerospace sector, while making aircraft skins, rivet holes are included. These are potential sources of crack initiation and fatigue failure and thus, having an understanding of SCFs would be helpful to minimize chances of an overall unsafe design. Similarly, in the biomedical industry, stress around screw holes of load-bearing implants must be carefully controlled to avoid unanticipated fracture or loosening, which could be fatal for patients using such devices. Lastly, another example emphasizing the importance of SCF would be in the civil engineering industry, where bridge beams may have significantly large holes for utilities. The stress at the rim of the holes could be significantly high, leading to sudden failure. Thus, in all engineering industries, underestimating or overestimating SCFs could lead to either unsafe designs or unnecessary waste of materials. Thus, it would be highly suggested to perform accurate modeling, that would be informed by mesh studies.

Lastly, to understand whether the results from the simulations are accurate enough, a Mesh Convergence Study was performed. A Mesh Convergence Study is a process of refining a finite element mesh, decreasing the size of the elements and seeing how parameters like temperature and stress change. The end goal is to reach a particular value after which changing element size has negligible change on the accuracy of the results. This is similar to what was performed in this lab wherein, different mesh sizes were investigated for each hole and the percentage difference was calculated in order to observe any noticeable trends. In general, finer meshes are better; however, they are expensive and require longer processing times. The key aim of this lab was thus to realize that the most optimal mesh is one that can provide convergence in the values being estimated, in this case, the value of K.

REFERENCE LIST

[1] M. Gonzalez. "Me 32301 - mechanics of materials laboratory." (2023).

[2] OpenAI. "ChatGPT (April 4th, 2025 Version)," [Online]. Available: ://openai.com/chatgpt (visited on 03/04/2025).

[3] R. Sengupta, Stress Concentration [Online]. Available: `https://web.njit.edu/~sengupta/met%20301/Stress%20Concentration.pdf`

[4] Dassault Systèmes, Abaqus Analysis User's Manual, Version 6.5, 2004. [Online]. Available: `https://classes.engineering.wustl.edu/2009/spring/mase5513/abaqus/docs/v6.5/books/gsa/default.htm?startat=ch04s04.html`

DATA SHEETS

The following Figures 10 and 11 are screenshots of the team's data collection spreadsheet. There are a total of 72 entries based on 8 holes and 9 mesh sizes. The entire spreadsheet was broken into two screenshots for better readability.

	A	B	C	D	E	F	G
1	2r/D	sig_norm [MPa	Mesh size [mm	sig_max [MPa]	K_FEA	K_Theo	Diff [%]
2	0.1	0.06172839506	1	0.15042	2.436804	2.721143	-10.44924872
3	0.1	0.06172839506	2	0.19038	3.084156	2.721143	13.34046024
4	0.1	0.06172839506	3	0.1272	2.06064	2.721143	-24.27299852
5	0.1	0.06172839506	4	0.16369	2.651778	2.721143	-2.549112634
6	0.1	0.06172839506	5	0.10664	1.727568	2.721143	-36.51314907
7	0.1	0.06172839506	6	0.14204	2.301048	2.721143	-15.43818168
8	0.1	0.06172839506	7	0.0937	1.51794	2.721143	-44.21682359
9	0.1	0.06172839506	8	0.13502	2.187324	2.721143	-19.61745487
10	0.1	0.06172839506	9	0.13136	2.128032	2.721143	-21.79639218
11	0.2	0.06944444444	1	0.18185	2.61864	2.506464	4.475468229
12	0.2	0.06944444444	2	0.19413	2.795472	2.506464	11.53050672
13	0.2	0.06944444444	3	0.18185	2.61864	2.506464	4.475468229
14	0.2	0.06944444444	4	0.18212	2.622528	2.506464	4.630587154
15	0.2	0.06944444444	5	0.17793	2.562192	2.506464	2.223371251
16	0.2	0.06944444444	6	0.17642	2.540448	2.506464	1.355854303
17	0.2	0.06944444444	7	0.16958	2.441952	2.506464	-2.573825118
18	0.2	0.06944444444	8	0.16636	2.395584	2.506464	-4.423761921
19	0.2	0.06944444444	9	0.1661	2.39184	2.506464	-4.5731357
20	0.3	0.07936507937	1	0.19868	2.503368	2.346801	6.671507299
21	0.3	0.07936507937	2	0.19947	2.513322	2.346801	7.095659155
22	0.3	0.07936507937	3	0.19858	2.502108	2.346801	6.61781719
23	0.3	0.07936507937	4	0.19915	2.50929	2.346801	6.923850808
24	0.3	0.07936507937	5	0.19861	2.502486	2.346801	6.633924223
25	0.3	0.07936507937	6	0.19716	2.484216	2.346801	5.855417652
26	0.3	0.07936507937	7	0.1937	2.44062	2.346801	3.997739902
27	0.3	0.07936507937	8	0.19082	2.404332	2.346801	2.451464781
28	0.3	0.07936507937	9	0.18911	2.382786	2.346801	1.533363928
29	0.4	0.09259259259	1	0.22805	2.46294	2.232992	10.29775297
30	0.4	0.09259259259	2	0.22945	2.47806	2.232992	10.97487138
31	0.4	0.09259259259	3	0.22777	2.459916	2.232992	10.16232929
32	0.4	0.09259259259	4	0.22754	2.457432	2.232992	10.05108841
33	0.4	0.09259259259	5	0.22687	2.450196	2.232992	9.727038879
34	0.4	0.09259259259	6	0.22669	2.448252	2.232992	9.639980797
35	0.4	0.09259259259	7	0.22497	2.429676	2.232992	8.808092461
36	0.4	0.09259259259	8	0.22109	2.387772	2.232992	6.931507144
37	0.4	0.09259259259	9	0.2182	2.35656	2.232992	5.533741276

Figure 5: Datasheet for 2r/D 0.1 to 0.4

1	2r/D	sig_norm [MPa	Mesh size [mm	sig_max [MPa]	K_FEA	K_Theo	Diff [%]
2	0.5	0.1111111111	1	0.27582	2.48238	2.155875	15.14489476
3	0.5	0.1111111111	2	0.28	2.52	2.155875	16.88989389
4	0.5	0.1111111111	3	0.2797	2.5173	2.155875	16.76465472
5	0.5	0.1111111111	4	0.27995	2.51955	2.155875	16.8690207
6	0.5	0.1111111111	5	0.28054	2.52486	2.155875	17.1153244
7	0.5	0.1111111111	6	0.27764	2.49876	2.155875	15.90467907
8	0.5	0.1111111111	7	0.27143	2.44287	2.155875	13.31222821
9	0.5	0.1111111111	8	0.27159	2.44431	2.155875	13.37902244
10	0.5	0.1111111111	9	0.27149	2.44341	2.155875	13.33727605
11	0.6	0.1388888889	1	0.36992	2.663424	2.106288	26.45108361
12	0.6	0.1388888889	2	0.37089	2.670408	2.106288	26.7826622
13	0.6	0.1388888889	3	0.37101	2.671272	2.106288	26.82368223
14	0.6	0.1388888889	4	0.36893	2.656296	2.106288	26.11266835
15	0.6	0.1388888889	5	0.36814	2.650608	2.106288	25.84261981
16	0.6	0.1388888889	6	0.36456	2.624832	2.106288	24.61885554
17	0.6	0.1388888889	7	0.36117	2.600424	2.106288	23.46003965
18	0.6	0.1388888889	8	0.36177	2.604744	2.106288	23.66513981
19	0.6	0.1388888889	9	0.3569	2.56968	2.106288	22.0004102
20	0.7	0.1851851852	1	0.52777	2.849958	2.075069	37.34280643
21	0.7	0.1851851852	2	0.54713	2.954502	2.075069	42.38090396
22	0.7	0.1851851852	3	0.54793	2.958822	2.075069	42.58908981
23	0.7	0.1851851852	4	0.55071	2.973834	2.075069	43.31253563
24	0.7	0.1851851852	5	0.53446	2.886084	2.075069	39.08376059
25	0.7	0.1851851852	6	0.53685	2.89899	2.075069	39.70571581
26	0.7	0.1851851852	7	0.53177	2.871558	2.075069	38.38373567
27	0.7	0.1851851852	8	0.51551	2.783754	2.075069	34.15235831
28	0.7	0.1851851852	9	0.53794	2.904876	2.075069	39.98936903
29	0.8	0.2777777778	1	1.0032	3.61152	2.053056	75.90947349
30	0.8	0.2777777778	2	1.0021	3.60756	2.053056	75.71659029
31	0.8	0.2777777778	3	0.99456	3.580416	2.053056	74.39446367
32	0.8	0.2777777778	4	0.98624	3.550464	2.053056	72.93556532
33	0.8	0.2777777778	5	0.97864	3.523104	2.053056	71.6029178
34	0.8	0.2777777778	6	0.97627	3.514572	2.053056	71.18734219
35	0.8	0.2777777778	7	0.95003	3.420108	2.053056	66.58620125
36	0.8	0.2777777778	8	0.88154	3.173544	2.053056	54.57659216
37	0.8	0.2777777778	9	0.9199	3.31164	2.053056	61.3029552

Figure 6: Datasheet for 2r/D 0.5 to 0.8

APPENDIX

```

clc          % Clean up Command Window
clear all    % Clean up all stored variables
close all    % Close all open windos

%set up colorlist
colorlist=["#6c3483","#CCCCFF", "#6495ED","#40E0D0","#9FE2BF",...
    "#DE3163","#FF7F50","#FFBF00","#f7dc6f","#2b7d38"];
```

```

%set up data
Filename="Lab8_Data";
Filetype='.csv';
data=readmatrix(append(Filename,Filetype));


%create sigma max figure
figure (1)
hold on
box on
grid minor
xlabel("Mesh Size [mm]")
ylabel("\sigma_{max} [MPa]")
ax = gca;
ax.FontName='Arial';
ax.FontSize=14;
xlim([1 12])

%code to set up sigma max and mesh vectors
for ii=1:8
    row_s = 1+9*(ii-1);
    row_f= 9+9*(ii-1);
    %increment vector:
    row = row_s:1:row_f;
    %x-values (mesh vec):
    mesh_vec = [data(row,3)];
    %y-values (sigma max):
    sigma_max_vec = [data(row,4)];
    %create plot
    plot(mesh_vec, sigma_max_vec, ":", "Marker", "*", "Linewidth", 2.5, ...
    "Color", colorlist(ii));
end
```

```

lgd = legend("2r/D=0.1", "2r/D=0.2", "2r/D=0.3", "2r/D=0.4", "2r/D=0.5", ...
    "2r/D=0.6", "2r/D=0.7", "2r/D=0.8", "location", "NorthEast", ...
    "fontsize", 11, 'Color', 'w');
exportgraphics(gcf, "sig_max.png");

%create K_FEA figure
figure (2)
hold on
box on
grid minor
xlabel("Mesh Size [mm]")
ylabel("K_{FEA}")
ax = gca;
ax.FontName='Arial';
ax.FontSize=14;
xlim([1 12])

for ii=1:8
    row_s = 1+9*(ii-1);
    row_f= 9+9*(ii-1);
    %increment vector:
    row = row_s:1:row_f;
    %x-values (mesh vec):
    mesh_vec = [data(row,3)];
    %y-values (K_fea):
    Kfea_max_vec = [data(row,5)];
    %create plot
    plot(mesh_vec, Kfea_max_vec, ":", "Marker", "*", "Linewidth", 2.5, ...
    "Color", colorlist(ii));
end

lgd = legend("2r/D=0.1", "2r/D=0.2", "2r/D=0.3", "2r/D=0.4", "2r/D=0.5", ...
```

```
    "2r/D=0.6", "2r/D=0.7", "2r/D=0.8", "location", "NorthEast", ...
    "fontsize", 11, 'Color', 'w');
exportgraphics(gcf, "K_fea.png");

%create K percent difference figure
figure (3)
hold on
box on
grid minor
xlabel("2r/D")
ylabel("K diff [%]")
ax = gca;
ax.FontName='Arial';
ax.FontSize=14;
xlim([.1 0.8])

for ii=1:9
    %increment vector:
    datalist = ii:9:72;
    %x-values (2r/D):
    rD_vec= [data(datalist,1)];
    %y-values (K_diff):
    K_diff_vec = abs(data(datalist,7));
    %create plot:
    plot(rD_vec, K_diff_vec, ":", "Marker", "*", "Linewidth", 2.5, ...
    "Color", colorlist(ii));
end

lgd = legend("1mm mesh", "2mm mesh", "3mm mesh", "4mm mesh", "5mm mesh", ...
    "6mm mesh","7mm mesh", "8mm mesh", "9mm mesh", "location", ...
    "NorthWest", "fontsize", 11, 'Color', 'w');
exportgraphics(gcf, "K_diff.png");
```